Dingrong Wang

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Professional Summary

Ph.D. Candidate in Computer Science with a stellar track record of publications at top-tier AI venues (CVPR, NeurIPS, KDD, ICLR, ICML). Specializing in Reinforcement Learning (RL/RLHF), advanced Computer Vision (Dense Object Detection, Test-Time Tuning), Multi-Modal Foundation Models (VLMs/LLMs), and Efficient Deep Learning (Network Compression, NAS, Sparsification). Proven capability in developing robust, real-time perception systems, designing closed-loop simulation strategies via RL, and optimizing large networks for deployment on edge constraints—highly applicable to Autonomous Vehicle (AV) tech stacks.

Education

Rochester Institute of Technology — 2020 Aug. – Present
PhD. Student in Computing Information and Science *GPA: 3.96* — *Rochester, NY*

Dalian University of Technology — 2016 Sep. – 2020 June.
Bachelor of Engineering (Digital Media Technology) *GPA: 3.87* — *Dalian, China*

Research Experience

Graduate Research Assistant — 08/2021-Present
Several topics, please refer to publication list for details — *Rochester, NY*

- Online/offline/off-policy reinforcement learning application to sketch retrieval, recommender system, dense objection detection, adversarial network architectural search/compression/pruning and time series analysis (classification/regression/forecasting).
- Imitation learning, inverse reinforcement learning, distributional RL, multi-agent RL, meta RL, DRL frameworks fine-tuned by few human feedback (RLHF).
- VLM/LLM/foundational multi-modal model related research, regarding domain generalization, test time prompt tuning, safety alignment controlled by RL agents.

Group member — 12/2018-01/2019
Vehicle obstacles detection based on binocular stereo vision technology — *Dalian, China*

- Selected the outstanding deep learning stereo vision algorithm, with KITTI data set to generate binocular stereo vision UV disparity map as the data sources.
- Read related papers, calculated U and V disparity maps through the UV disparity map, and then restored the U and V disparity maps to the original picture in order to mark the obstacles.

Research Assistant — 06/2017-07/2017
Design and Implementation of Content-Based Near-Duplicate Chart Retrieval System — *Dalian, China*

- Learned to utilize the most popular algorithm of image features extraction, such as perceptual hash algorithm and edge detection histogram algorithm to build algorithm models
- Utilized web crawler techniques to collect pictures and charts as data sources and stored them into the SQL database.
- Designed the software framework, and used a series of frames and tools to realize the connection and coupling with front end, data base and algorithm
- Verified the robustness of algorithm through images transformation, such as stretching, expanding and shrinking, based on Python image processing technology

Industry Experience

Member of Technical Staff — 09/2019-05/2020
Chinese OCR and full stack website development — *Ruikebang Technology Co., Beijing, China*

- Used Chinese OCR yolo+CRNN model for text detection and scanning
- Tried image processing algorithms such as Hough transform and sifted algorithm to and detect lines and characters.
- Front-end website maintenance using JavaScript, CSS and HTML.
- Develop back-end machine learning algorithms to analyze data.
- Develop queries using SQL within Java DAO to retrieve data from database.

Applied Scientist Intern — 06/2024-09/2024
LLM adaptation, prompt engineering, time series forecasting — *Amazon, Bellevue, WA*

- Construct large-scale ASIN data set with Amazon million-size data set retrieved from AWS service
- Prompt engineering and fine-tuning LLMs (llama, Claude, falcon) into the downstream tasks
- Leverage Prophet, Saison to conduct time series decomposition and forecasting

Applied Scientist Intern — 05/2025-08/2025
LLM global reasoning and domain adaptation, knowledge graph based information retrieval — *Amazon, San Diego, CA*

- GraphRAG based LLM reasoning and information retrieval
- LLM evaluation benchmark build up
- Prompt engineering, reinforcement learning assisted RAG optimization.

Technical Skill

Machine Learning

- **Deep Learning Framework:** TensorFlow, PyTorch
- **Machine Learning and Data Analysis Library:** Scikit-learn, Pandas, MatplotLib
- **Object Detection Framework:** OpenCV, Detection2, MM-detection
- **RL, VLM related:** HuggingFace, LLaVA, DPO, PPO, GRPO

Software Programming

- **Tools:** C, C++, Java, Python, R, Matlab, SQL, Tableau, Qt, Unix, CUDA, HTML, CSS, JavaScript
- **Courses:** Data Structure, Parallel Computing, Image Processing

Publications

Coupling Deep Textural and Shape Features for Sketch Recognition. PROCEEDINGS OF THE 28TH ACM INTERNATIONAL CONFERENCE ON MULTIMEDIA
Qi Jia, Xin Fan, Meiyu Yu, Yuqing Liu, Dingrong Wang, Longin Jan Latecki

Deep Reinforced Attention Regression for Partial Sketch Based Image Retrieval. PROCEEDINGS OF THE 21TH IEEE INTERNATIONAL CONFERENCE ON DATA MINING
Dingrong Wang, Hitesh Sapkota, Xumin Liu, Qi Yu

Deep Temporal Sets with Evidential Reinforced Attentions for Unique Behavioral Pattern Discovery. ICML 2023
Dingrong Wang, Deep Shankar Pandey, Qi Yu

Distributionally Robust Ensemble of Lottery Tickets Towards Calibrated Sparse Network Training. Neurips 2023
Hitesh Sapkota, Dingrong Wang, Qi Yu

LIBR+: Improving Intraoperative Liver Registration by Learning the Residual of Biomechanics-Based Deformable Registration. MICCAI 2024
Dingrong Wang, Soheil Azadvar, Jon Heiselman, Xiajun Jiang, Michael Miga, Linwei Wang

Reinforced Compressive Neural Architecture Search for Versatile Adversarial Robustness. KDD 2024
Dingrong Wang, Hitesh Sapkota, ZHIQIANG TAO, Qi Yu

Adaptive Important Region Selection with Reinforced Hierarchical Searching for Dense Object Detection. Neurips 2024
Dingrong Wang, Hitesh Sapkota, Qi Yu

Conservative Evidential Exploration of Long-Term User Interest in Dynamic Recommender Systems. ICLR 2025
Dingrong Wang, Krishna Prasad Neupane, Qi Yu

Inferring online retailing product seasonality via LLMs for cold-start cases. Computing Conference 2026
Dingrong Wang, Zhenyu Zhang (Amazon), Jie Zhou (Amazon), Muhammad Tayyab Asif (Amazon)

Accessible Learning Labs: Accessibility Education Through Experiential Learning 2024 36th International Conference on Software Engineering Education and Training (CSEE&T)
Y Liu, X Que, D Wang, SA Malachowsky, DE Krutz

InsCal: Calibrated Multi-Source Fully Test-Time Prompt Tuning for Object Detection. CVPR 2026
Xiaofan Que, Dingrong Wang, Daniel Krutz, Qi Yu